LECTURE 3: FLOWCHARTS

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AGENDA

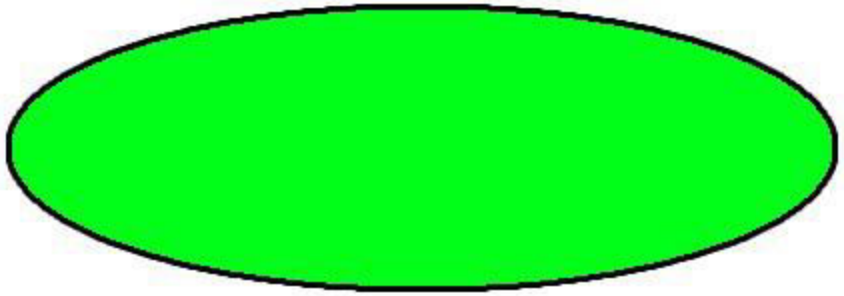
- Flow Charts
- Flow Charts Vs Algorithm
- Phases of Compiler
- Data types

FLOW CHARTS

- Flowchart is a graphical representation of an algorithm.
- Programmers often use it as a program-planning tool to solve a problem.
- It makes use of symbols which are connected among them to indicate the flow of information and processing.
- The process of drawing a flowchart for an algorithm is known as “flowcharting”.

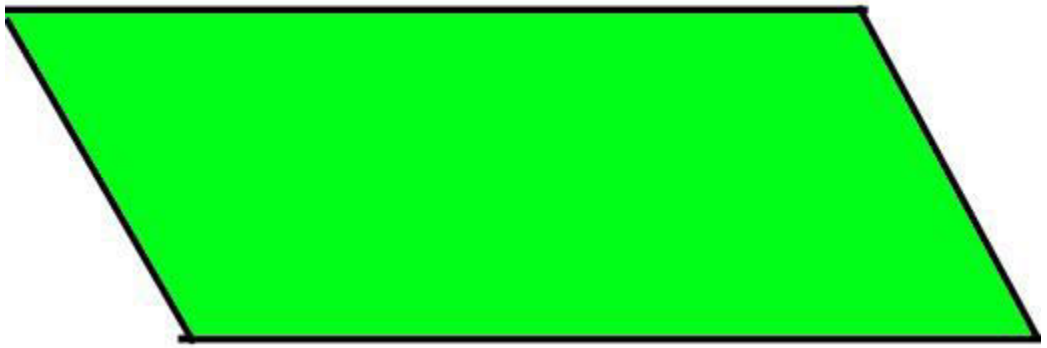
BASIC SYMBOLS

- **Terminal:** The oval symbol indicates Start, Stop and Halt in a program's logic flow. A pause/halt is generally used in a program logic under some error conditions. Terminal is the first and last symbols in the flowch



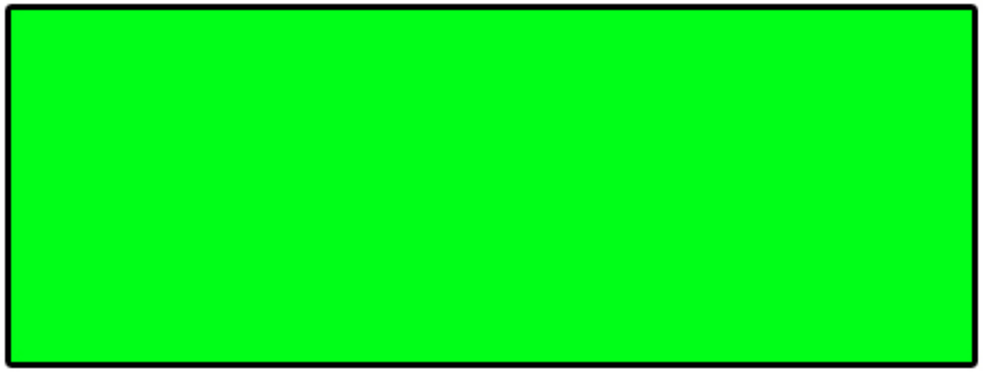
SYMBOLS

- **Input/Output:** A parallelogram denotes any function of input/output type. Program instructions that take input from input devices and display output on output devices are indicated with parallelogram in a flowchart.



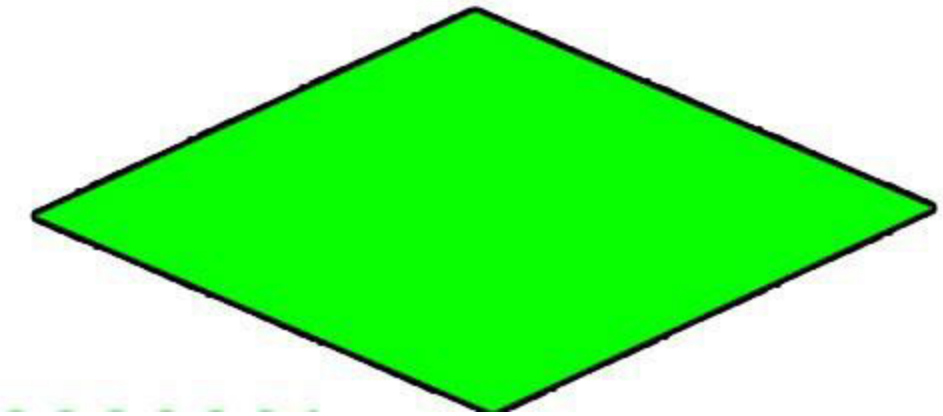
SYMBOLS

- **Processing:** A box represents arithmetic instructions. All arithmetic processes such as adding, subtracting, multiplication and division are indicated by action or process symbol.



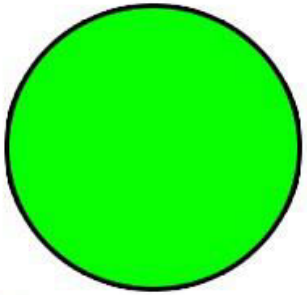
SYMBOLS

- **Decision** Diamond symbol represents a decision point. Decision based operations such as yes/no question or true/false are indicated by diamond in flowchart.



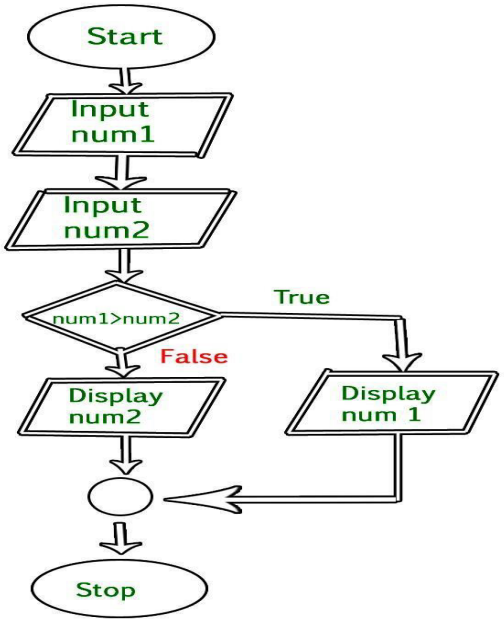
SYMBOLS

- **Connectors:** Whenever flowchart becomes complex or it spreads over more than one page, it is useful to use connectors to avoid any confusions. It is represented by a circle.



EXAMPLE

- Draw a flowchart to input two numbers from user and display the largest of two numbers



FLOW CHARTS VS ALGORITHMS

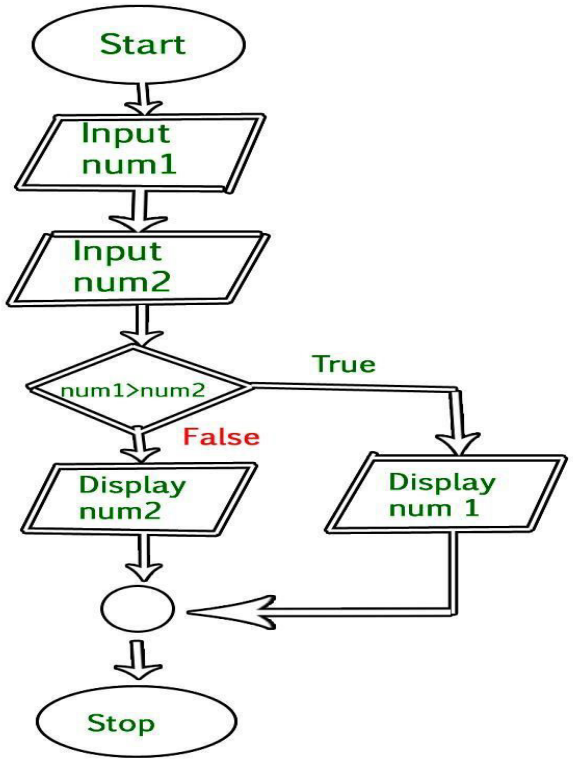
- Flowcharts are more easier to understand than algorithms as they are represented in pictorial forms.
- One can easily write the code by looking at the flowchart.
- Yet **algorithms are still preferred** and here are the reasons in next Slide.

FLOW CHARTS VS ALGORITHMS

- One can write an algorithm even for the complex codes whereas it becomes difficult in a flowchart.
- Conditional and Looping statements can be written in algorithms but these statements are difficult to present in flowcharts.
- Debugging for an algorithm is easier whereas a flowchart cannot be debugged as it is quite complex to do so.
- There are certain rules to be followed in case of flowcharts but there are no particular rules for algorithms.

ALGORITHM

- ***Step 1:*** *Begin*
  Step 2: *read a, b;*
  Step 3: *check if(a>b)*
  Step 4: *display a;*
  Step 5: *else*
  Step 6: *display b;*
  Step 7: *End*



PHASES OF COMPILER

- Lexical Analysis(Lexemes)
- Syntax Analysis(Parse tree)
- Semantic Analysis
- Intermediate Code generation
- Code Optimization
- Code Generation
- Symbol Table

DATA TYPES

- **Primitive Data Types**: These data types are built-in or predefined data types and can be used directly by the user to declare variables. example: int, char , float, bool etc. Primitive data types available in C++ are:
- Integer
- Character
- Boolean
- Floating Point
- Double Floating Point
- Void
- Wide Character

DATA TYPES

- We will focus on four different Primitive types :
- bool is a boolean value (true / false) ;
- int is an integer value (−2147483648 to 2147483647) ;
- double is a floating-point value (precision goes from 2.2250738585072014× 10−308 to 1.7976931348623157× 10308) ;
- char is a character (letter, digit, punctuation, etc.)

DATA TYPES

- **Derived Data Types:** The data-types that are derived from the primitive or built-in datatypes are referred to as Derived Data Types. These can be of four types namely:
- Function
- Array
- Pointer
- Reference

DATA TYPES

- **Abstract or User-Defined Data Types**: These data types are defined by user itself. Like, defining a class in C++ or a structure. C++ provides the following user-defined datatypes:
- Class
- Structure
- Union
- Enumeration
- Typedef defined DataType


Thank
you!!!